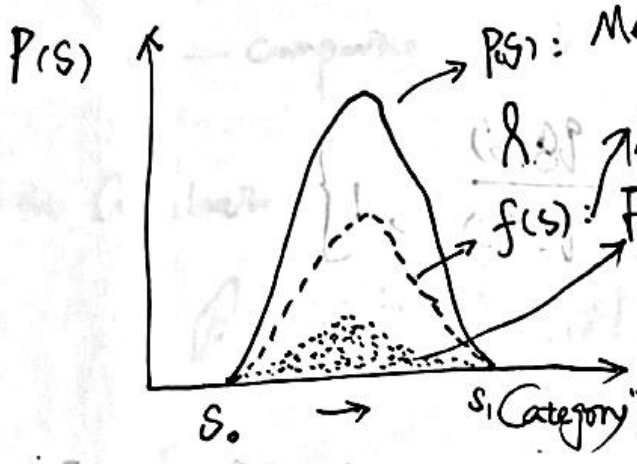
Sanborn. 2010. Uncovering mental ~ MCMC



$P(S)$: Mental representation's distribution (Not known mostly)

δ : Non-zero indicators ~~of samples from~~ $P(S)$

$f(S)$: Frequency of samples (MCMC) from $P(S)$ (approximation to $P(S)$)

Category "S" – representation "S" ⇒ objects "S_0, S_1..."

$$P(S) \propto f(S)$$

approximate ← Samples → generate (MCMC)

MCMC – generating samples from arbitrary distribution.

$$\underbrace{\overset{\text{initial value}}{A} \xrightarrow{\text{transition probabilities } P(B|A)} B \to C \to D \cdots \to D \to \cdots}_{\text{Stationary distributed states}} \xrightarrow{\text{determine}} \text{Target distribution } P(S)$$

※ Metropolis

Transition prob. < Proposal distribution. $q(S'|S)$; Acceptance function. $a(S';S)$ >

Detailed Balance: $$\underbrace{P(S)\underbrace{q(S'|S)\,a(S';S)}_{P(S\to S')} = P(S')\,q(S|S')\,a(S;S')}_{P(S)\nearrow,\ P(S\to S')\searrow} \Rightarrow$$

$$\Rightarrow \frac{P(S)}{P(S')} = \frac{q(S|S')\,a(S;S')}{q(S'|S)\,a(S';S)}$$

with $q(S|S') = q(S'|S)$ <symmetric proposal>, $\dfrac{P(S)}{P(S')} = \dfrac{a(S;S')}{a(S';S)}$

$$\Rightarrow a(S';S) = \min\left\{\frac{P(S')}{P(S)}, 1\right\} \quad {}^{*}\left(\frac{P(S')}{P(S)} = \frac{f(S')}{f(S)}\right) \checkmark$$

※ Hasting accepted function

$$a(S';S) = \min\left\{\frac{P(S')\ q(S|S')}{P(S)\ q(S'|S)}, 1\right\}$$

※ Barker AF: with symmetric proposal: $q(\sim) = q(\sim)$

$$a(S';S) = \frac{P(S')}{P(S')+P(S)}$$

Human decision: act as an $a(S';S)$ in MCMC.

$$a(S';S) = \frac{f(S')^r}{f(S')^r + f(S)^r}$$

$P(S) \propto f(S)^r$ MCMC $\Rightarrow$ Human prior distribution

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$$P(c|S) \propto \overbrace{P(S|c)}^{\text{Structural Knowledge of a category}} P(c) \approx \frac{P(S|c)}{P(S|c)+P(S'|c)}$$

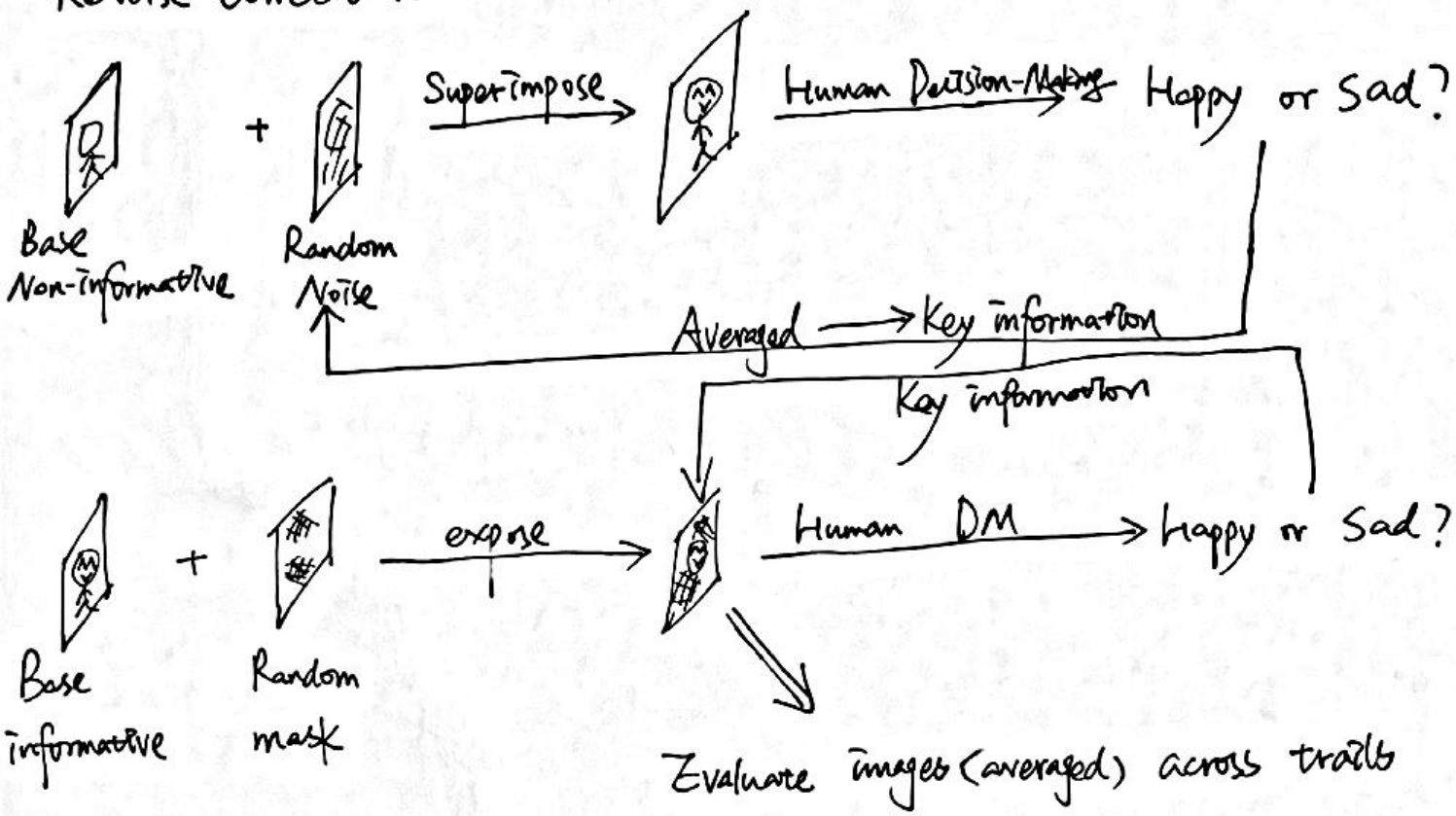
$P(S|c) \propto f(S;c)$ = Simlarity between S and Category C

So, $a = \underbrace{\dfrac{f(S';c)}{f(S';c)+f(S;c)}}_{\text{Barker AF}}$ $\rightarrow$ MCMC samples from $P(S|c)$

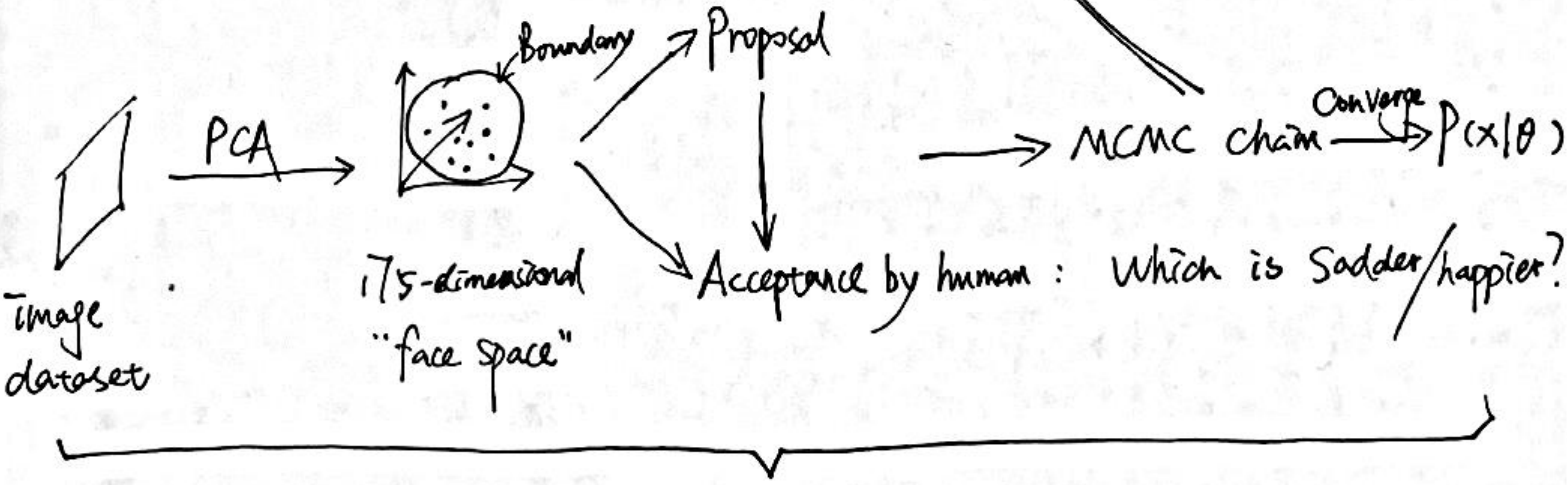
2011, testing MCMCP using facial affect categories.

— Comparing RC with MCMCP

Reverse Correlation:



MCMCP:



Stimuli $\xrightarrow{\text{parameterization}}$ High-dimensional continuous space in which stimuli embed $\xrightarrow{\text{draw samples}}$ MCMCP $\rightarrow P(x|\theta)$

Artificial images,

Realistic images with PCA,

...

Sanborn. 2015 Extensions of MCMCP

- Discrete MCMCP

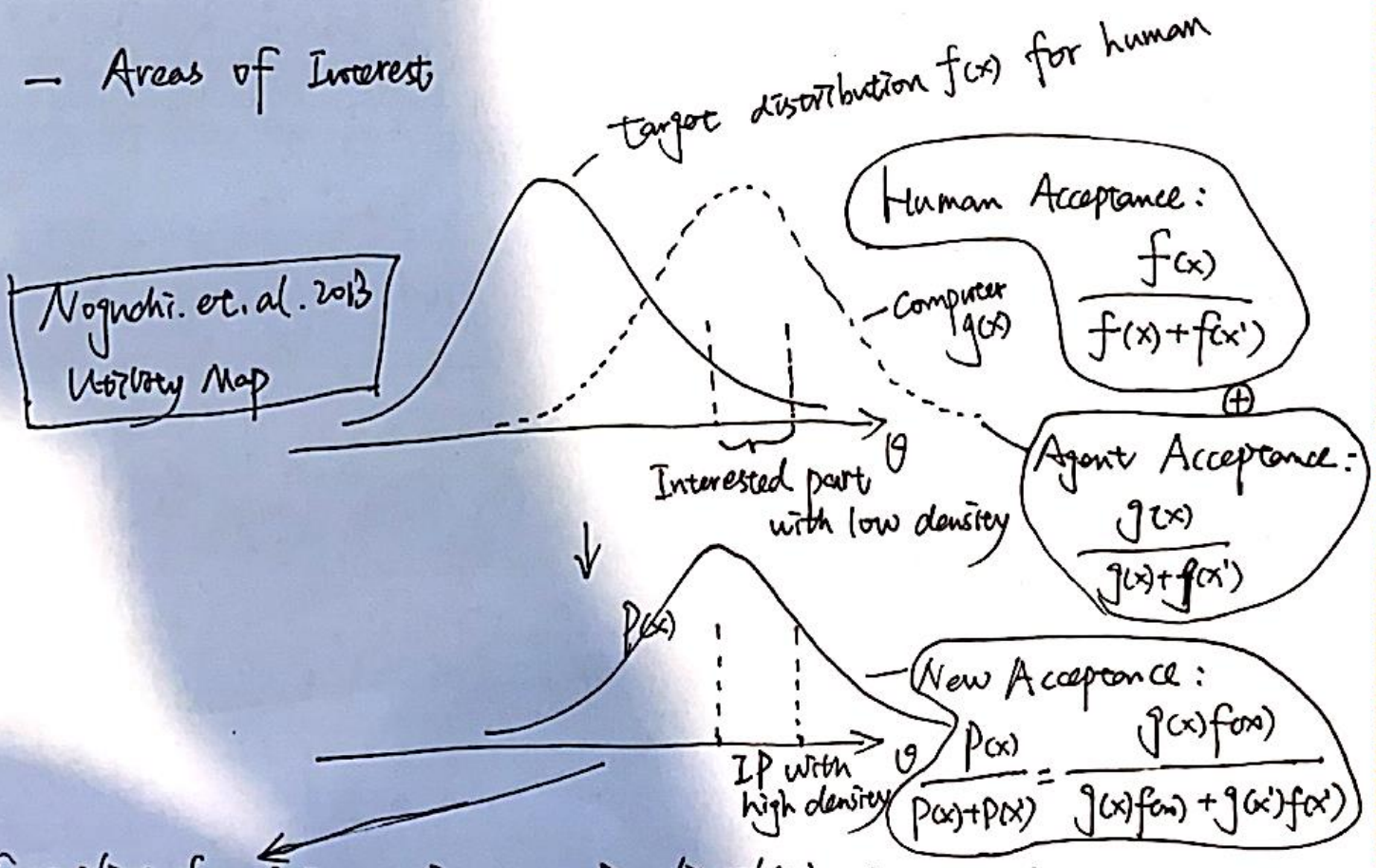
dataset

$\xrightarrow[\text{e.g. color, PCA, ...}]{\text{Quantify}}$ Pair-wised similarity Matrices $\xrightarrow{\text{b-matching}}$ Graph consisted of discrete stimuli $\xrightarrow{\text{draw samples}}$ MCMCP...

$$\arg\max_G \sum_{i,j} G_{ij} S_{ij}, \quad \sum_i G_{ij} = b \text{ (constant)}; \; G_{ii}=0; \; G_{ij}=G_{ji}$$

G : adjacency matrix of the graph, "$G_{ij}=1$" means an edge

S : similarity matrix

—— Hsu. et. al. 2019

- Areas of Interest